File Matching with Faulty Continuous Matching Variables

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CES 17-45 **June, 2017**

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Abstract

We present LFCMV, a Bayesian file linking methodology designed to link records using continuous matching variables in situations where we do not expect values of these matching variables to agree exactly across matched pairs. The method involves a linking model for the distance between the matching variables of records in one file and the matching variables of their linked records in the second. This linking model is conditional on a vector indicating the links. We specify a mixture model for the distance component of the linking model, as this latent structure allows the distance between matching variables in linked pairs to vary across types of linked pairs. Finally, we specify a model for the linking vector. We describe the Gibbs sampling algorithm for sampling from the posterior distribution of this linkage model and use artificial data to illustrate model performance. We also introduce a linking application using public survey information and data from the U.S. Census of Manufactures and use LFCMV to link the records.

Keyword*:* latent class; faults; record linkage; file linking; energy efficiency

[*]This work was supported by NSF Grant SES 1131897, and by the Duke University Energy Initiative Energy Research Seed Fund, with co-funding from the Information Initiative at Duke. Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no con dential information is disclosed.

1 Introduction

In the energy economics literature, there is interest in examining the "energy-efficiency gap", or failure to implement cost-effective technologies to improve energy efficiency (Allcott and Greenstone, 2012; Gerarden et al., 2015). Lack of information about cost-effective technologies and practices is commonly cited as a market failure resulting in the energy efficiency gap. Some government policies aim at filling the information gap to enable improvements in energy efficiency. The U.S. Department of Energy Industrial Assessment Center (IAC) program is one such policy intervention, providing no-cost efficiency assessments to small and medium enterprises (SME). Assessments are conducted by teams of university faculty and students who visit manufacturing plants to assess productivity, energy use and waste. Visits result in a report detailing all cost-saving opportunities identified during the assessment (U.S. Department of Energy, 2016).

While the IAC program has assembled a wealth of data, the SME self-select their participation and there is little information about the SME after the assessment. In short, the IAC data do not include information about non-participants or about participants after the assessment. This reality limits the research into the energy efficiency gap that can be conducted using IAC data. One way to extend the research potential of the IAC database is to link the data to a longitudinal data base that includes pre- and post-assessment informations. Ideally, this longitudinal data base would also include information on plants that did not receive IAC assessments, facilitating the comparison of plants receiving assessments to a peer group of plants that did not.

Such longitudinal plant-level data is available in the U.S. Census of Manufactures (CMF). The CMF is a census containing information on every U.S. manufacturing plant. In addition to information on sales, production, and product information, the CMF contains information related to energy usage. Manufacturing plants that receive IAC energy assessments represent a subset of the plants in the United States, and hence a subset of the plants in the CMF. Linking a record in the IAC to a record in the CMF is synonymous with declaring that the manufacturing plant associated with the Census record received an IAC energy assessment. Such a link would provide information on the energy efficiency recommendations provided to a plant by IAC, enabling researchers to examine CMF energy

data within the context of the IAC recommendations.

Records from the IAC assessments are publicly available. However, due to privacy restrictions, identifying information such as plant name have been removed from the released IAC records. Linking IAC records to CMF records therefore relies on a set of categorical and continuous variables common to both files. However, linking in this scenario is complicated by a few characteristics of the data. First, the number of IAC assessments conducted in a given year is much smaller than the number of CMF records available for the same year. This means that linking on common categorical variables rarely results in unique matches in the CMF. Second, the continuous variables common to both files do not necessarily agree for true matches. For instance, values in the IAC relating to sales information appear to be rounded, and information for each completed form is provided at potentially different times of the year. This means that even for true matches, we do not expect values of common continuous variables to agree exactly.

To facilitate linking, we develop a linking methodology that incorporates these challenges into the modeling structure and accounts for some uncertainty in the matching process. The method utilizes common categorical variables, which we call blocking variables (BVs), to define groups, known as blocks, of records in the larger file that contain possible matches for each record in the smaller file. This version of blocking is a common file linking technique that restricts the number of possible matches for any given record by requiring that any potential matches agree exactly on the BVs. In contrast to the work in Dalzell and Reiter (2017), we assume all variables used for blocking are known to be fault-free. Once blocks of records are identified, we utilize common continuous variables, which we call matching variables (MVs), to estimate which in-block record in the larger file is a match for each record in the smaller. We assume that these MVs may not agree exactly across matched pairs. We refer to the entire model as LFCMV, which stands for linking with faulty continuous matching variables.

LFCMV linking proceeds in four main steps. First, we group records into blocks based on patterns of agreement in the BVs. Second, we utilize a linking model that models the distance between the MVs of records in the smaller file and the MVs of their matching records in the larger file. This linking model is conditional on a vector C that indicates

which record in the larger file is matched to each record in the smaller. Third, we specify a mixture model for the distance component of the linking model. This latent structure allows the distance between MVs in matched pairs to vary across types of matched pairs. Finally, within each block defined by the BVs, we specify a model for the corresponding elements of C.

The remainder of this paper is organized as follows. In Section 2, we introduce the LFCMV methodology. In Section 3, we describe the Gibbs sampling steps used to obtain posterior estimates of the linkage structure. In Section 4, we use simulation studies to illustrate the performance of LFCMV. In Section 5, we present results of additional studies that investigate the performance of LFCMV under a variety of fault scenarios. In Section 6, we apply the model to link records from the 2007/2008 IAC assessments to the 2007 CMF. We conclude in Section 7.

2 Model

In this section, we describe the LFCMV methodology. Notation and key concepts are presented Section 2. We then present the model for C in Section 2.2, and the linking model in Section 2.3.

2.1 Notation

Let the two files to be linked be denoted F_1 and F_2, containing n_1 and n_2 records, respectively. Without loss of generality, we assume that $n_1 < n_2$. We further assume that each record $i \in F_1$ has a match $i^{'} \in F_2$, that is, the set of individuals in F_1 is a subset of the set of individuals in F_2. This assumption is motivated by the application in which the IAC assessment records (F_1) represent a subset of the records in the CMF (F_2). We assume each record corresponds to a single true individual.

Our goal is to link each record $i \in F_1$ to a record $i^{'} \in F_2$. To reduce the number of possible matches for each i, we place each of the n_1 records in F_1 in its own block. Let (i) denote the block defined by $i \in F_1$. For each record i, we limit the possible matches to a set of records $i^{'} \in F_2$ which are assigned to block (i). Possible matches $i^{'} \in F_2$ are assigned

to (i) as follows.

Let J denote the number of BVs, i.e., the number of categorical variables common to F_1 and F_2 that are used for blocking. It is not necessary to use all common categorical variables for blocking; analysts can select variables appropriate for blocking for each application. As blocking restricts possible matches to record pairs with identical values of the BVs, BVs should be selected in such a way that the variable values are expected to agree across matched pairs. Let B_{fik} denote the BV value for record i in file f on field k, where $k = 1, \ldots, J$, $i = 1, \ldots, n_f$, and $f = 1, 2$. For all (f, i), let $B_{fi} = (B_{fi1}, \ldots, B_{fiJ})$. For each $i \in F_1$, define $F_{2(i)}$ as the set of all $i^{'} \in F_2$ such that $B_{1ik} = B_{2i^{'}k}$ for all $k = 1, \ldots, J$. Assign all records in $F_{2(i)}$ to block (i). In other words, for each i, we restrict the set of possible matches to records $i^{'}$ in F_2 with the same BV values as record i.

The BVs available in the CMF and IAC are the state in which a plant is located and the 6-digit NAICS code (U.S. Census Bureau, 2016) corresponding to the specific products made by the plant. We require that all possible matches agree on both BVs. This is a reasonable first step because we expect that if $(i, i^{'})$ is a match, values of state and NAICS should agree across files. As an example, consider an IAC record for a plant in Oregon with NAICS code 342156. The block of possible matches, or block, for this record then contains all CMF records corresponding to plants in Oregon with the NAICS code 342156.

Under the LFCMV blocking structure, it is possible that a record $i^{'}$ in F_2 can be considered a possible match for more than one record i in F_1. Each record i in F_1 is placed in its own block, and the set of possible matches $F_{2(i)}$ are defined only by the BV combination B_{fi}. If two records a and b in F_1 have the same BV combination, their sets of possible matches, $F_{2(a)}$ and $F_{2(b)}$, contain the same records from F_2. This yields the possibility that both a and b can be linked to the same record in F_2. For our application, this is not a concern. Our goal is to identify records in the CMF that have received an IAC assessment, not necessarily to generate a set of matched pairs. If unique matches are required for a given application, one could imagine adapting this blocking structure to suit this need. For instance, blocks may be defined such that more than one F_1 record may be assigned to a block. Matching would then assign each record in $F_{2(i)}$ to at most one record i in F_1.

With n_1 records in F_1, we have n_1 blocks of records from F_2. Each block $F_{2(i)}$ represents potential matches for the accompanying record $i \in F_1$. Let $n_{(i)}$ be the number of records in $F_{2(i)}$. For each $i \in F_1$, the goal of file linking is to determine which of the $n_{(i)}$ records is a match for i. We define $C_i = i^{'}$ such that $i^{'} \in F_{2(i)}$ and $(i, i^{'})$ is a match. The n_1 element vector $C = (C_1, \ldots, C_{n_1})$ then specifies which records from F_2 match the records in F_1.

Finally, let p be the number of common continuous variables (MVs) used for matching. For linking the IAC and CMF, $p = 2$ continuous variables, sales (TVS) and number of employees (TE), serve as the MVs. TVS is a continuous variable for the sales of each plant in U.S. dollars, and TE refers to the total number of employees for the plant. For all $j = 1, \ldots, p$, let Y_{ij} denote the standardized value for MV j for record i in F_1. We provide details on this standardization in Section 4.1. For all $i = 1, \ldots, n_1$, let $Y_i = (Y_{i1}, \ldots, Y_{ip})$. Similarly, let $\mathbf{Y}_j = (Y_{ij}, \ldots, Y_{n_1 j})$ for all j. We define $\mathbf{Y}$ as the $n_1 \times p$ matrix of MV values for F_1. Similarly, for all j, let $X_{i^{'}j}$ denote the standardized value for MV j for record $i^{'} \in F_2$ where $i^{'} = 1, \ldots, n_2$. Let $X_i = (X_{i1}, \ldots, X_{ip})$ and $\mathbf{X}_j = (X_{ij}, \ldots, X_{n_1 j})$. We define $\mathbf{X}$ as the $n_2 \times p$ matrix of MV values for F_2.

2.2 Model for C

Within a block $F_{2(i)}$, there are $n_{(i)}$ possible values of C_i. We denote these values as ℓ, where $\ell \in \{1, \ldots, n_{(i)}\}$. We assume that *a priori*, each $i^{'}$ in $F_{2(i)}$ is equally likely to be linked with i, so that for all ℓ,

$$p(C_i = \ell \mid B_{1i}) = \frac{1}{n_{(i)}}. \tag{1}$$

Conditional on B_{1i}, which defines the block for record i, each C_i is modeled independently. Similar modeling structures are used in Gutman et al. (2013).

2.3 Linking model

Conditional on C, we assume that for each matching pair $(i, i^{'} = C_i)$, the values of Y_i are related to X_{C_i} through a linking model. This model reflects the belief that for true matches $(i, i^{'} = C_i)$, we may have $Y_{ij} \neq X_{i^{'}j}$ for some $j = 1, \ldots, p$. Such a discrepancy could occur due to incorrect recording, time differences in data collection, different accuracy

thresholds, etc. The notion of linking based on examining distances between variables common to both files is an example of distance-based record linkage. Distance-based record linkage is commonly used for re-identification purposes in disclosure and privacy research (e.g., Pagliuca and Seri, 1999; Domingo-Ferrer and Torra, 2002a, 2003; Torra et al., 2006). In these applications, distances between each record i and $i^{'}$ are computed using some distance metric, and the record pair that has the smallest distance according to this metric is considered a link (Domingo-Ferrer and Torra, 2002b).

We model Y_i with a multivariate normal distribution centering Y_i at its corresponding X_{C_i}. This choice of linking model favors matched pairs with similar values of Y and X. The variance component of the linking model represents the distance between Y_i and X_{C_i} that is considered plausible for matched pairs. In our application, each IAC record corresponds to a manufacturing plant. It is reasonable to assume that the distance in the MVs across matched pairs may vary with state, plant type, or other characteristics of the records. In more general applications, the distance may vary across certain blocks, or with other features in the data. To allow the linking model to capture these distributional features, we utilize a mixture of multivariate normals. Mixtures of multivariate normals are highly flexible and, with enough components, can represent any distribution. For LFCMV, the mixture framework allows the distance across types of matched pairs to vary with latent class.

Following standard conventions for a mixture model, assume each pair $(i, i^{'} = C_i)$ belongs to one of H latent classes. For convenience, we associate the latent class with record i. Let $z_i \in \{1, \ldots, H\}$ represent the latent class assignment for $i \in F_1$, with $z = (z_1, \ldots, z_{n_1})$. For $h = 1, \ldots, H$, we assume that $Pr(z_i = h) = \pi_h$ for all i. Let $\pi = (\pi_1, \ldots, \pi_H)$.

Conditional on z and C, we model Y_i with a multivariate normal distribution centered at X_{C_i} and class-specific variance Σ_h. This mixture model can be written as

$$Y_i|\mathbf{X}, C, \mathbf{\Sigma}, z_i \sim N_p\left(X_{C_i}, \Sigma_{z_i}\right) \tag{2}$$

$$z_i|\boldsymbol{\pi} \sim Multinomial\left(1; \pi_1, ..., \pi_H\right). \tag{3}$$

This is an adaption of the standard mixture of multivariate normals in which each Y_i is modeled as $Y_i \mid \boldsymbol{\mu}, \mathbf{\Sigma}, z \sim N_p\left(\mu_{z_i}, \Sigma_{z_i}\right)$. In the standard formulation, each Y_i is assumed

to have a component specific mean μ_{z_i}. In our framework, we center each Y_i at its corresponding X_{C_i}. This reflects a belief that for matched pairs, Y_i and X_{C_i} should be close together. When linking the IAC and CMF data, $p = 2$, so (2) is bivariate normal.

Following Kim et al. (2014), we select a conjugate prior structure for each Σ_h as follows:

$$\Sigma_h \sim InvWishart\left(f_0, G_0\right), \tag{4}$$

where $G_0 = Diag(\phi_1, ..., \phi_p)$ and

$$\phi_j \sim Gamma(a_\phi, b_\phi), \tag{5}$$

with $E(\phi_j) = a_\phi / b_\phi$. This conjugate prior structure facilitates posterior updates, as discussed in Section 3. In order to ensure a proper posterior distribution, we set $f_0 = p + 1$. We set the hyper-parameters $a_\phi = b_\phi = 0.25$, a choice which allows substantial prior mass at modest sized variances (Kim et al., 2014).

For the latent class weights π, we use a stick-breaking representation of the truncated Dirichlet process (Sethuraman, 1994; Ishwaran and James, 2001). In this framework, the mixture probabilities are

$$\pi_h = V_h \prod_{g<h} (1 - V_g), h = 1, ..., H \tag{6}$$

$$V_h \sim Beta(1, \alpha), V_H = 1 \tag{7}$$

$$\alpha \sim Gamma(a_\alpha, b_\alpha). \tag{8}$$

We set $a_\alpha = b_\alpha = 0.25$, reflecting a low prior sample size and hence a vague prior for α. As noted in Escobar and West (1995), such a choice allows the data to dominate in the posterior.

The stick-breaking representation allows the data to inform the number of latent classes that are occupied at any iteration of the posterior sampler. As H represents the maximum number of occupied classes, we recommend starting with a large value, such as $H = 30$. When posterior runs indicate that all H classes are consistently occupied, we increase H and restart the sampling.

There are a few considerations to take into account when using LFCMV for file linking. First, as is true for file linking methodologies in general, block size is important to the

performance of the model. Smaller blocks tend to lead to a higher match rate, while for extremely large blocks, the model often selects a match essentially at random. It is therefore important to select a blocking scheme that results in small blocks. Second, the linking model described in Section 2.3 is based on distance between Y_i and its corresponding X_{C_i}. As we illustrate in Section 4, in the posterior, the model tends to assign high probability to matches with small distances across matched pairs. The determination of what is a reasonable distance across a matched pair is influenced by the class-specific variance component of the linking model.

3 Posterior sampling

In this section, we describe the Gibbs sampler used to estimate the posterior distribution of the LFCMV model.

3.1 Initialization

We initialize $C^{(0)}$ such that

$$C_i^{(0)} = \left(i^{'} \mid d(i, i^{'}) = min\left(d(i, \ell), \ell \in F_{2(i)} \right) \right), \tag{9}$$

where

$$d(i, i^{'}) = \sqrt{\sum_{j=1}^{p} \left(\frac{Y_{ij} - X_{i^{'}j}}{Y_{ij}} \right)^2}. \tag{10}$$

In other words, within a block, we initialize C_i such that $d(i, i^{'})$ is minimized. The two closest records according to this metric will be initialized as links. Because Y_i and X_{C_i} are standardized variables, $d(i, i^{'})$ assigns equal weight to the distances in each of the p MVs. The importance of such standardization in distance-based linkage is discussed in Pagliuca and Seri (1999).

For the linking model, we set $\alpha^{(0)} = 1$, and initialize V, π, z, and $\mathbf{\Sigma}$ by drawing from the appropriate distributions from Section 2.3.

3.2 Posterior sampling algorithm

With the truncated representation of the Dirichlet process, posterior sampling for the entire model is facilitated with a Gibbs Sampler as follows.

1. For $h = 1, ..., H$, update Σ_h from the full conditional

$$\Sigma_h^{(s+1)} | \mathbf{Y}, \mathbf{X}, C^{(s)}, z^{(s)} \sim InvWishart\left(f_0 + n_h^{(s)}, G_0 + S_h^{(s)}\right), \tag{11}$$

where $n_h^{(s)} = \sum_i I(z_i^{(s)} = h)$ denotes the number of individuals in latent class h at iteration (s) and $S_h^{(s)} = \sum_{i:[z_i^{(s)}=h]} (Y_i - X_{C_i^{(s)}})(Y_i - X_{C_i^{(s)}})^T$.

2. For $h \in \{1, ..., H-1\}$, update V_h from the full conditional,

$$V_h^{(s+1)} | z^{(s)}, \alpha^{(s)} \sim Beta\left(1 + n_h^{(s)}, \alpha^{(s)} + \sum_{g=h+1}^{H} n_g^{(s)}\right). \tag{12}$$

Set $V_H^{(s+1)} = 1$.

3. Set $\pi^{(s+1)} = V_h^{(s+1)} \prod_{g<h} (1 - V_g^{(s+1)})$ for all $h \in \{1, ..., H\}$ per (6).

4. For $j = 1, \ldots, p$, update ϕ_j from the full conditional

$$\phi_j^{(s+1)} | \mathbf{\Sigma}^{(s+1)} \sim Gamma\left(a_\phi + \frac{1}{2} H f_0, b_\phi + \frac{1}{2} \sum_{h=1}^{H} \left(\Sigma_h^{-1}[j,j]^{(s+1)}\right)\right), \tag{13}$$

where $\left(\Sigma_h^{-1}[j,j]\right)^{(s+1)}$ represents the jth diagonal entry of $(\Sigma_h^{-1})^{(s+1)}$.

5. Update α from the full conditional,

$$\alpha^{(s+1)} | \pi^{(s+1)} \sim Gamma\left(a_\alpha + H - 1, b_\alpha - log(\pi_H^{(s+1)})\right). \tag{14}$$

6. For $i \in 1, ..., n$, sample the latent class indicator $z_i \in \{1, ..., H\}$ from a multinomial full conditional,

$$z_i^{(s+1)} | \phi^{(s+1)}, C^{(s)} \sim Multinomial({\pi_1}^*, {\pi_2}^*, ..., {\pi_H}^*), \tag{15}$$

where

$${\pi_h}^* = \frac{\pi_h^{(s+1)} f(Y_i | X_{C_i^{(s)}}, \Sigma_h^{(s+1)})}{\sum_{g=1}^{H} \pi_g^{(s+1)} f(Y_i | X_{C_i^{(s)}}, \Sigma_g^{(s+1)})}. \tag{16}$$

7. For $i = 1, ..., n_1$, we update each C_i. For each $i^{'} \in F_{2(i)}$,

$$Pr\left(C_i = i^{'} | Y_i, \mathbf{X}, z_i^{(s+1)}, \mathbf{\Sigma}^{(s+1)}\right) = \frac{f(Y_i|\mathbf{X}, C_i = i^{'}, \Sigma_h^{(s+1)})}{\sum\limits_{\ell \in F_{2(i)}} f(Y_i|\mathbf{X}, C_i = \ell, \Sigma_h^{(s+1)})}. \tag{17}$$

The update in (17) represents draws from the multinomial Bernoulli posterior distribution of C. Because each block contains only one Y_i, computing (17) requires enumerating $n_1 \times \left(\sum\limits_i n_{(i)}\right)$ multivariate normal likelihoods at each MCMC iteration. Unlike many file linking applications in which there are multiple records from each file in each block, for LFCMV this direct updating is fairly efficient in terms of computation time.

4 Illustrative simulation

In this section we illustrate the performance of the LFCMV model before applying it to the IAC and CMF model in Section 6. For this simulation, we use IAC data to create both F_1 and F_2. F_1 and the corresponding matches are constructed using records from 2007 and 2008, while the remainder of F_2 is constructed using records from the remaining IAC years. In Section 4.1, we describe the process of creating F_1 and F_2 and introduce the BVs and MVs used in linking. Simulation results are presented in Section 4.2.

4.1 Data

The complete downloaded IAC database (up through the year 2016) contains 17583 records with 56 variables. In this simulation, as well as in the creation of the linked data product described in Section 6, we use two categorical variables, NAICS code and state, as the BVs. NAICS codes are nested up to the 4-digit level, meaning that all plants that produce a certain type of product, say dairy products, must agree on at least the first 4 digits of their NAICS codes. In this simulation we define blocks using state and the first 4-digits of the 6-digit NAICS code.

Two continuous variables, sales and number of employees, serve as our MVs. Sales is a continuous variable for the sales of each plant in U.S. dollars, and employees refers to the total number of employees for the plant. We let $j = 1$ refer to sales and $j = 2$ to number of employees.

Table 1: Creating F_1 and F_{2U}. Left column: describes a data cleaning step necessary before linking. Center column: the number of records in F_1 after each cleaning step is performed. Right column: the number of records in F_{2U} after each cleaning step. The final row gives the number of records in the final subsets F_1 and F_{2U} used for linking.

Action	F_1 Records	F_{2U} Records
Initialize	All 2007/2008: 797	All other years: 16786
Remove records from Puerto Rico	786	16779
Remove States Unique to F_1 or F_2	778	16440
Remove Missing States	778	13622
Remove Missing Sales	764	13476
Remove Missing Employees	764	13464
Remove Missing Electricity Usage	763	13429
Remove Missing Cost of Energy	763	13422
Require Matching State/4-Digit NAICS	512	1170
Final Total:	512	1170

4.1.1 Create F_1

To create F_1, we use the IAC records from 2007 and 2008, comprising 797 records. We reduce this set to the final set of 512 F_1 records using the steps outlined in the second column of Table 1. For this simulation, we use use 16786 IAC records from years other than 2007 or 2008 to create possible matches for F_1. If a record in F_1 has a state and 4-digit NAICS combination which is not found in these 16786 records, we remove these records from consideration for F_1. For simulation purposes, we also exclude from F_1 any records containing missing data. The process of incorporating missing data imputation into the file linking process adds an additional level of uncertainty. The selection process for F_1 results in the final count of $n_1 = 512$ records.

4.1.2 Create F_2

We create F_2 in two stages. First, for each record i in F_1, we create a matching record by adding noise to the MVs in record i. The collection of these n_1 matches is denoted F_{2M}. Second, we use the 16786 IAC records from years other than 2007 or 2008 to create sets of possible matches for each record i in F_1. These records represent the false matches, or possible options for i based on state and NAICS agreement which are in fact not a match for i. The collection of these records is denoted F_{2U}. We let $F_2 = (F_{2U}, F_{2M})$.

We create F_{2M} as follows. For each record i in F_1, we generate a matching record using the following steps. Assign each record i in F_1 to one of $H = 30$ latent classes z_i with

$$z_i \sim Multinomial(1; 1/H, \ldots, 1/H). \tag{18}$$

To generate the variance matrix, we sample from the following model:

$$\Sigma_h \sim InvWishart\left(50, G_{0h}\right), \tag{19}$$

where $G_{0h} = Diag(\phi_{1h}, \phi_{2h})$ and

$$\phi_{1h} \sim Gamma(1,1), \phi_{2h} \sim Gamma(25,3), \tag{20}$$

with $E(\phi_j) = a_\phi / b_\phi$. Let $\mathbf{Y}^{(0)}$ be the $n_1 \times p$ matrix containing all non-standardized MV values in F_1. For each i with latent class $z_i = h$, we define

$$F_{2M,i} = Y_i^{(0)} + \epsilon_i, \epsilon_i \sim N_2((0,0), \Sigma_{z_i}). \tag{21}$$

Here $F_{2M,i}$, $Y_i^{(0)}$ and ϵ_i are vectors of length $p = 2$. We repeat this match generating process 100 times for each i, creating 100 replicates $F_{2M,i}^{(m)}$, $m = 1, \ldots, 100$. Let $F_{2M}^{(m)} = \{F_{2M,i}^{(m)} \mid i = 1, \ldots n_1\}$.

We create F_{2U} using the 16786 IAC records from years other than 2007 or 2008. After removing missing data and other records as outlined in Table 1, we select 1170 records which have a state/4-digit NAICS combination that is observed in the F_1 data. Denote these 1170 records as F_{2U}. We let $F_2^{(m)} = (F_{2U}, F_{2M}^{(m)})$, where $m = 1, \ldots, 100$. For each replicate m, we link F_1 and $F_2^{(m)}$.

Each of the n_1 records from F_1 is assigned to a block (i). Based on the BVs, we assign each of the 1170 records in F_{2U} to these blocks. For each record $i \in F_1$, let $F_{2U(i)}$ denote

Table 2: Block sizes for the illustrative simulation. Block Size: the number of records from F_{2U} assigned to each block (i). Count: the number of blocks of each block size.

Block Size	2	3	4	5	6	7	8	9	10	11	12	13	14
Count	129	106	76	39	20	17	30	14	11	23	4	4	10
Block Size	16	17	18	19	24	25	26						
Count	2	3	4	4	8	3	5						

the subset of F_{2U} assigned to block (i). As seen in Table 2, this results in blocks which range in size from 2 to 26. Blocks sizes, F_{2U}, and F_1 are consistent across replicates.

4.1.3 Standardization of Y and X

Before linking F_1 and $F_2^{(m)}$, we log transform the MVs. Let $\tilde{\mathbf{Y}}_j$ refer to the log-transformed MV values for field j in F_1, and $\tilde{\mathbf{X}}_j^{(m)}$ refer to the log-transformed MV values for field j in $F_2^{(m)}$. For ease of posterior computation in the linking model, we standardize $\tilde{\mathbf{Y}}_j$ and $\tilde{\mathbf{X}}_j^{(m)}$ as follows. Denote $S^{(m)} = (\tilde{\mathbf{Y}}_1, \tilde{\mathbf{X}}_1^{(m)})$ and $E^{(m)} = (\tilde{\mathbf{Y}}_2, \tilde{\mathbf{X}}_2^{(m)})$. Let $\bar{E}^{(m)}$ be the mean of all $n_1 + n_2$ elements of $E^{(m)}$ and let $\bar{S}^{(m)}$ be the mean of all $n_1 + n_2$ elements of $S^{(m)}$, with $sd(E^{(m)})$ and $sd(S^{(m)})$ representing the standard deviations of $E^{(m)}$ and $S^{(m)}$, respectively. We then standardize each transformed value as

$$\mathbf{Y}_1^{(m)} = (\tilde{\mathbf{Y}}_1 - \bar{S}^{(m)})/sd(S^{(m)}),\ \mathbf{X}_1^{(m)} = (\tilde{\mathbf{X}}_1^{(m)} - \bar{S}^{(m)})/sd(S^{(m)}), \tag{22}$$

$$\mathbf{Y}_2^{(m)} = (\tilde{\mathbf{Y}}_2 - \bar{E^{(m)}})/sd(E^{(m)}),\ \mathbf{X}_2^{(m)} = (\tilde{\mathbf{X}}_2^{(m)} - \bar{E}^{(m)})/sd(E^{(m)}). \tag{23}$$

4.2 Results

For each of 100 replicates, we apply the LFCMV model to link F_1 and the replicate specific $F_2^{(m)}$. We run the Gibbs sampler described in Section 3.2 for 5000 iterations with a burn in length of 200 iterations. For purposes of comparison, we also link each pair of data sets using a "naive" matching method. Specifically, for the naive method, we select a match $i^{'} = C_i$ such that

$$C_i = \left(i^{'} \mid d(i, i^{'}) = min\left(d(i, \ell), \ell \in F_{2(i)}\right)\right), \tag{24}$$

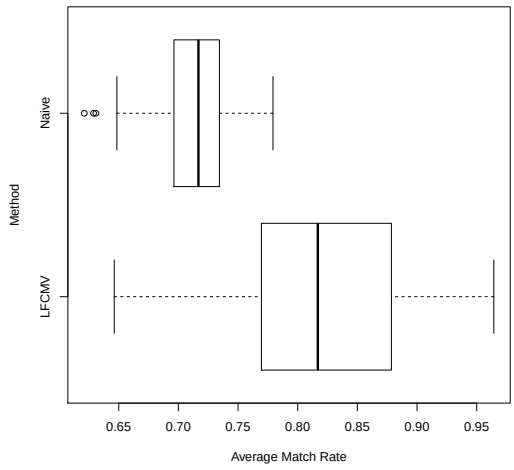


Figure 1: Average match rates for 100 simulations. Upper Boxplot: Match rates obtained by linking records by the naive method. Lower Boxplot: Average match rate for the LFCMV runs.

where

$$d(i, i') = \sqrt{\sum_{j=1}^{p} \left(\frac{Y_{ij} - X_{i'j}}{Y_{ij}} \right)^2}. \tag{25}$$

The primary metric used in this simulation is the match rate (MR), or the percentage of the n_1 records from F_1 that are correctly matched. For each replicate, the naive matching method yields one estimate of C. Conditional on this C, we compute a single estimate of the match rate. As a Bayesian procedure, LFCMV yields a set of posterior draws for C. For each replicate, we compute the MR for each posterior draw of C. We then average these MR values to obtain one estimate of the posterior MR for each replicate.

Results are displayed in Figure 1. For all but 2 of the 100 replicates, the MR obtained using LFCMV is higher than the MR obtained using the naive approach. A 95% posterior interval for the improvement in MR using LFCMV over the naive method is (.2%, 22%) with an average increase in match rate of 9%. This translates to an average of 47 additional records that are correctly matched using LFCMV.

As seen in Figure 1, the range of MR values is fairly wide. This range is due to the process of generating the X_i values. The average distance across matched pairs, i.e., the

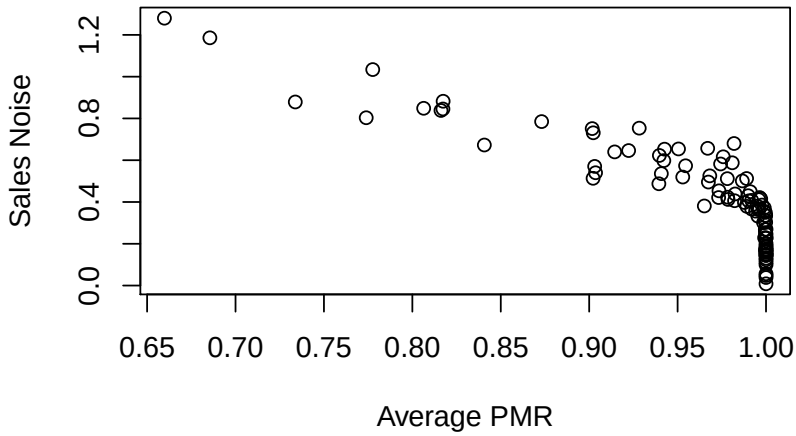


Figure 2: Average posterior match rates by average distance across matches pairs for sales from the 100 replicates. Lower PMR values are associated with greater distance.

average value of $Y_{ij} - X_{C_i j}$, is different across each replicate. Consider Figure 2. Replicates with smaller average PMR have, on average, more distance between the MV values for sales across matched pairs. By design, LFCMV is searching for a match with small distance between Y_i and X_{C_i}. If the data generation process creates a matching X_{C_i} which is far from Y_i, the model sometimes selects records $X_{i' \neq C_i}$ with a smaller distance between Y and X. This behavior is consistent with the intuition discussed in Section 2.3.

5 Sensitivity to distance across matched pairs

In this section, we illustrate the performance of LFCMV linkage when the mechanism that introduces distance across matched pairs of MVs is not normally distributed. The creation of F_1 and F_{2U} is the same as the simulation study in Section 4. However, we vary the process of creating F_{2M} by using different distributions to create the distance across matched pairs. In Section 5, we apply uniform distance at three levels: 10%, 20% and 30%. For this simulation, we are interested in examining the performance of LFCMV with increasing distances between matches pairs. In Section 5.2, we introduce distance

across matched pairs dependent upon the number of employees in a plant. This simulation leverages the ability of LFCMV to allow the distance between matched pairs to vary across types of pairs.

5.1 Uniform distance

In Section 4.2, the MR obtained by LFCMV tends to vary with the distance across matched pairs. To further explore this concept, we conduct a simulation in which distance between matched Y and X is generated uniformly at three different levels. Specifically, for each replicate, we generate $F_{2M}^{(m)}$ using

$$F_{2M,i1}^{(m)} = Y_{i1}^{(0)}(1+\kappa_{ij}), \kappa_{ij} \sim Uniform(0,u) \tag{26}$$

$$F_{2M,i2}^{(m)} = Y_{i2}^{(0)}(1+\kappa_{ij}), \kappa_{ij} \sim Uniform(0,u), \tag{27}$$

where $u = .1, .2$, and $.3$. We run 100 replicates at each setting of u.

Figure 3 displays the results across all three settings of u. As is true in Section 4, LFCMV generally yields higher match rates than the naive approach, and this result holds across all three settings of u. As u increases and true matched pairs have a greater distance between the MVs of matched pairs, both the naive and LFCMV linking methods have reduced performance. As previously mentioned, block size also has an impact on MR. Table 3 shows the breakdown of posterior MR by block size. The average PMRs for smaller blocks tend to be higher than average PMR for larger blocks, with some variation attributable to the number of blocks of a given size.

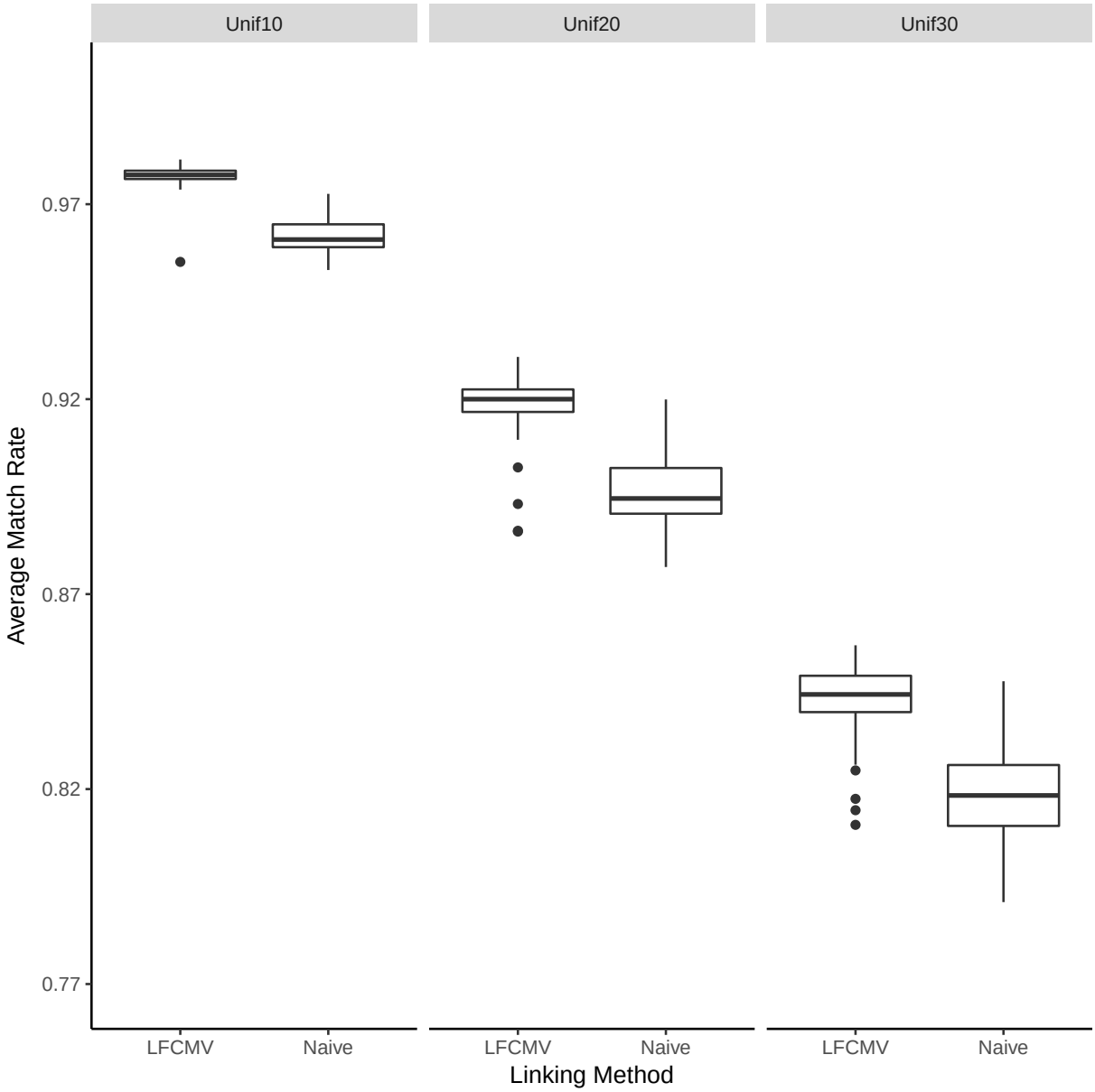


Figure 3: Match rates under uniform distance. Left Boxplot in each panel: Composed of average PMR for each of 100 replicates. Right Boxplot in each panel: Match rate for each replicate obtained by linking records by the "naive" approach, i.e., minimizing $d(i, i^{'})$.

Table 3: Average posterior match rate for each block size. The first column indicates the size of the block, with the number of blocks of that size in parentheses. Block sizes are consistent for each simulation. Remaining columns denotes the average posterior MR, as computed by averaging the block-specific MR across each of the 100 replicates.

Block Size (Count)	10%	20%	30%
2 (*129*)	.99	.98	.94
3 (*106*)	.99	.97	.93
4 (*76*)	.99	.94	.88
5 (*39*)	.98	.92	.80
6 (*20*)	.97	.90	.82
7 (*17*)	.98	.90	.78
8 (*30*)	.95	.89	.80
9 (*14*)	.91	.78	.66
10 (*11*)	.82	.56	.41
11 (*23*)	.93	.79	.64
12 (*4*)	1	1	1
13 (*4*)	.97	.85	.67
14 (*10*)	.99	.89	.72
16 (*2*)	1	.80	.57
17 (*3*)	1	.98	.82
18 (*4*)	1	.89	.61
19 (*4*)	.91	.50	.30
24 (*8*)	.99	.69	.40
25 (*3*)	1	.83	.57
26 (*5*)	1	.95	.80
Overall PMR	.98 (.003)	.92 (.007)	.84 (.008)

Table 4: Average posterior match rate for the SD and LD simulations. Bold values indicate the PMR associated with the plant size with smaller distance between matched pairs in each simulation. Large plants: the PMR for large plants, averaged over the 100 replicates of the SD and LD simulations, respectively. Small plants: the PMR for small plants, averaged over the 100 replicates of the SD and LD simulations, respectively. Overall PMR: the PMR for all plants, averaged over the 100 replicates of the SD and LD simulations, respectively. Values in parentheses represent the standard error.

	SD	LD
Large plants	**.90**(.008)	.78(.01)
Small plants	.76(.009)	**.88**(.003)
Overall PMR	.85(.006)	.84(.006)

5.2 Size dependent distance

One of the features of LFCMV is the ability to model distances between the MVs that vary with types of matched pairs. To illustrate this capability, we conduct a simulation in which we introduce correlation between the number of employees in a plant and the distance across matched pairs of MVs. We classify the 261 plants in F_1 with more than 130 employees as "large" and classify the remaining 251 plants in F_1 with 130 or fewer employees as "small". We run a simulation, denoted LD, in which MVs for matched pairs corresponding to large plants tend to be farther apart than the MVs for matched pairs corresponding to small plants. We then run a second simulation, denoted SD, in which the opposite is true. For simulation LD, we generate $F_{2M}^{(m)}$ from

$$F_{2M,i,j}^{(m)} = Y_{ij}(1+\kappa_{ij}), \kappa_{ij} \overset{iid}{\sim} \begin{cases} Uniform(0,.4) & : \text{ if } Y_{i2}^{(0)} > 130 \\ Uniform(0,.15) & : \text{ otherwise.} \end{cases} \tag{28}$$

Similarly, for simulation SD, we generate $F_{2M}^{(m)}$ from,

$$F_{2M,i,j}^{(m)} = Y_{ij}(1+\kappa_{ij}), \kappa_{ij} \overset{iid}{\sim} \begin{cases} Uniform(0,.4) & : \text{ if } Y_{i2}^{(0)} \leq 130 \\ Uniform(0,.15) & : \text{ otherwise.} \end{cases} \tag{29}$$

We run 100 replicates of the LD and SD simulation settings.

With these simulations, we are interested in assessing two aspects of LFCMV. First, we examine the match rate for different types of matched pairs for both the SD and LD simulations. The matching results are summarized in Table 4. As with the results from Section 4, larger distance across matched pairs tends to be associated with lower MRs than smaller distance across matched pairs. However, an examination of the match rate by plant size reveals that the naive approach and LFCMV perform differently in the different size groups. Consider Figure 4, displaying the match rates for the LD replicates. For large plants, which have greater distance across matched pairs in this simulation, LFCMV matches more accurately than the naive approach. However, the opposite trend is evident for smaller plants. This suggests that LFCMV overestimates the distance reasonable for matched pairs in the smaller plants, resulting in a lower match rate. Figure 5 shows similar tendencies for the SD simulation; LFCMV's match rate is better on average for the smaller plants, with the naive approach matching more accurately for the larger plants with less distance across matched pairs. Despite this behavior, LFCMV results in an overall improvement in match rate over the naive approach in both the SD and LD replicates.

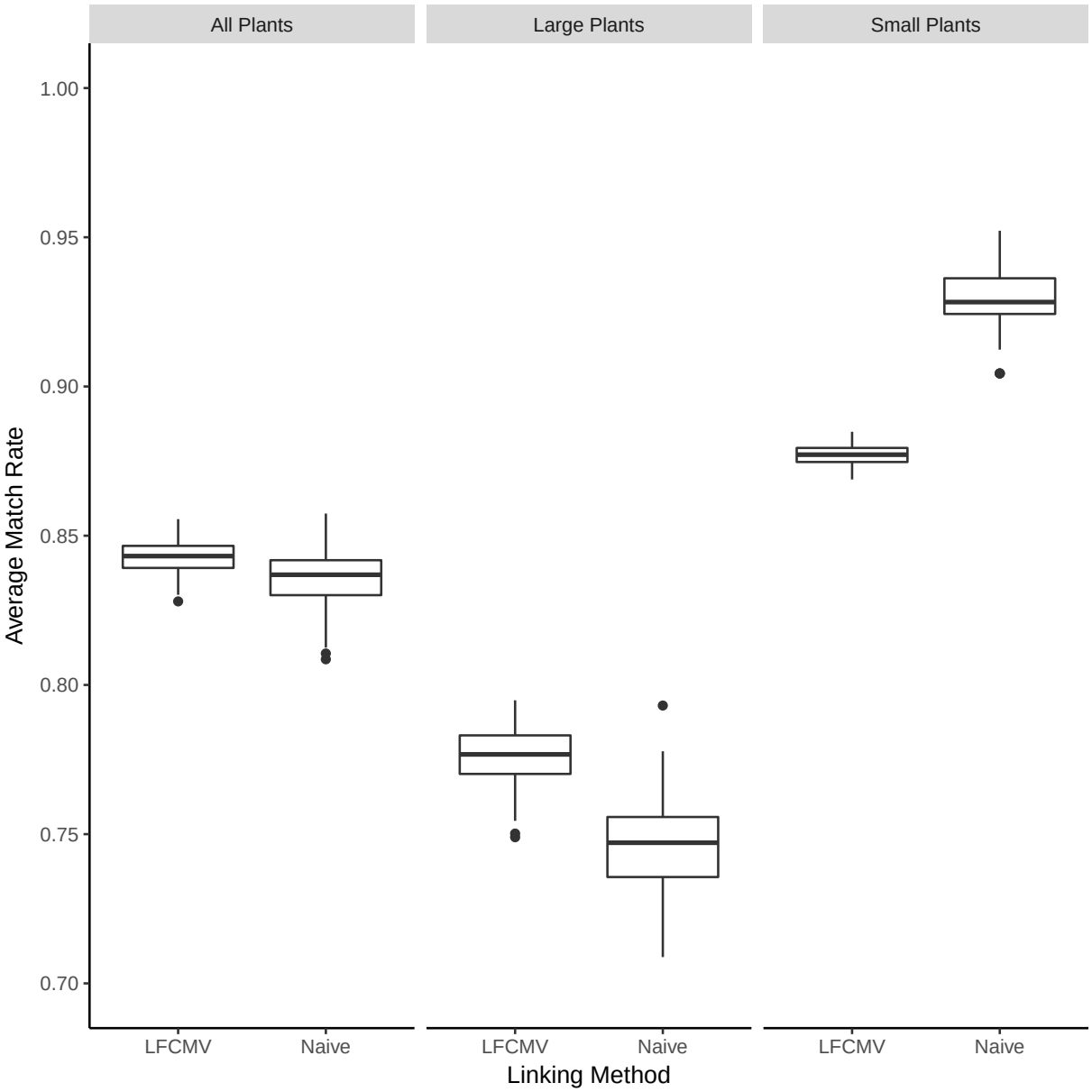


Figure 4: LD: Match rates under when large plants have greater distance across matched pairs than small plants. Left Boxplot in each panel: Composed of average PMR for each of 100 replicates. Right Boxplot in each panel: Match rate for each replicate obtained by linking records by the "naive" approach, i.e., minimizing $d(i, i^{'})$.

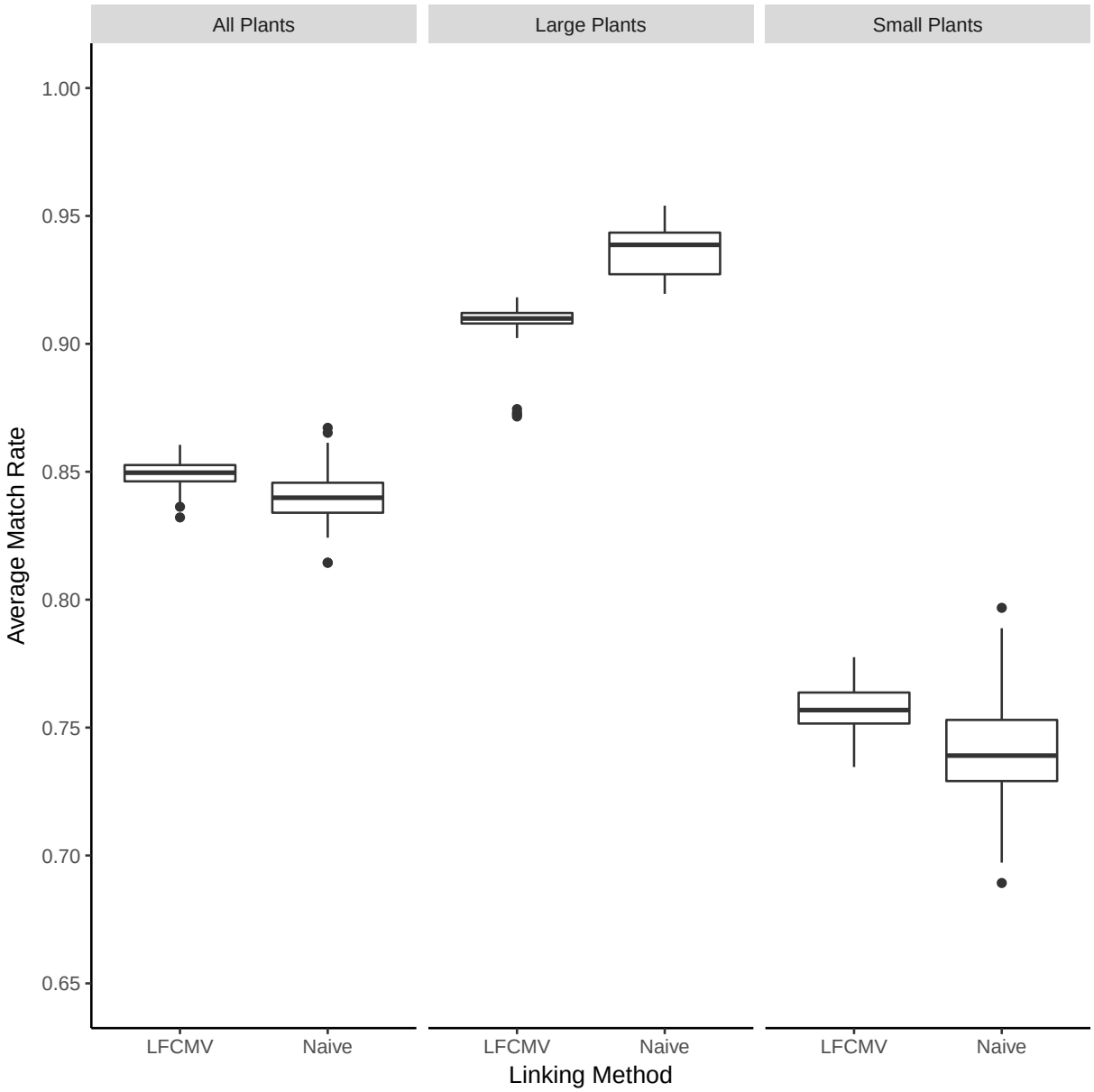
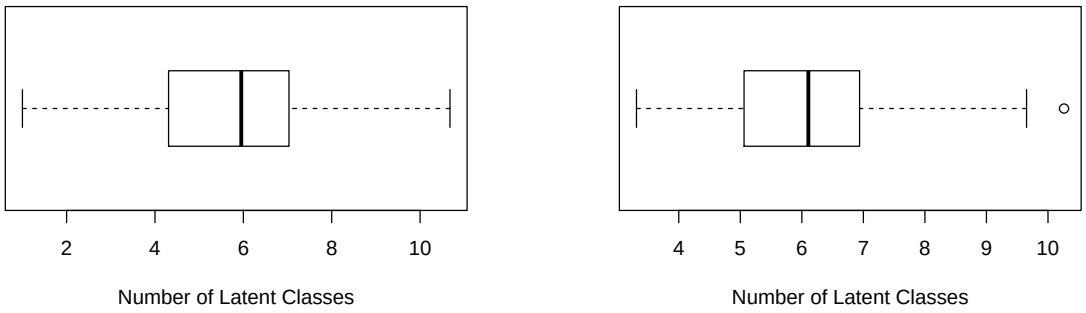


Figure 5: SD: Match rates under when small plants have larger distance across matched pairs than larger plants. Left Boxplot in each panel: Composed of average PMR for each of 100 replicates. Right Boxplot in each panel: Match rate for each replicate obtained by linking records by the "naive" approach, i.e., minimizing $d(i, i^{'})$.



(a) SD (b) LD

Figure 6: Average number of latent classes for the SD and LD simulations. For each of the 100 replicates of simulation SD (Figure 6a) and LD (Figure 6b), we compute the number of latent classes at each posterior iteration with more than 5 records. We then average these counts to obtain a single estimate of the average number of occupied latent classes for each replicate. The boxplots are composed of the 100 replicate-specific counts.

The second aspect of LFCMV we illustrate with the SD and LD simulations is the ability of LFCMV to capture the relationships inherent in the different types of matched pairs using the mixture model formulation of the linking model. In the simulation studies described in Section 5, the distance across matched pairs is uniform, so latent classes are not needed to describe the distribution of distances across matched pairs. Accordingly, the number of occupied mixture components collapses to a single component during the posterior sampling. However, for the SD and LD simulations, there are two clearly defined types of matched pairs, as for each simulation, small and large plants are associated with a certain degree of distance across matches. Accordingly, Figure 6 indicates that multiple mixture components are occupied in the SD and LD simulations. In Figure 7, we examine these components for a single iteration of a replicate of the SD simulation. The figure indicates that latent class 1 is associated with smaller plants than latent class 2, while latent class 3 contains a blend of plant sizes. As distance across matched pairs is directly related to plant size, these results suggest that the latent class structure of the linking model may be useful for capturing features in the data associated with differing amounts

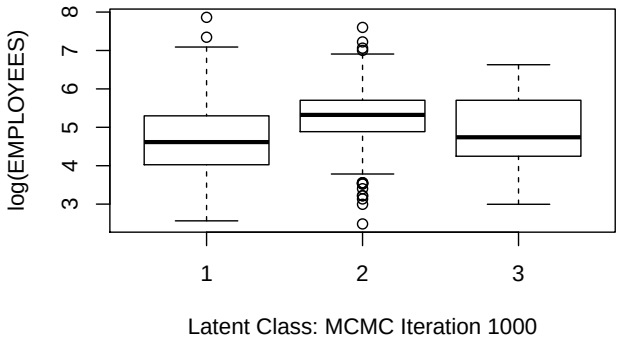


Figure 7: SD: The box plots display the log(employees) value for the three latent classes at MCMC iteration 1000. Each boxplot represents a latent class, and the y-axis represents the value of $\tilde{\mathbf{X}}_2$, denoted log(employees), for each record in that latent class. A paired t-test comparing the log(employees) value of latent classes 1 and 2 leads statistically significant results ($p < 0.05$). Specifically, latent class 1 tends to contain smaller plants than latent class 2.

of distance across matched pairs.

5.3 Summary of findings

The simulations conducted in Sections 5 and 5.2 highlight a few features of LFCMV linking. First, as is evident in Section 4.2, large distances among the MVs for matched pairs can lead to a reduction in PMR. This underscores the importance of the considerations discussed in Section 2.3. Second, the simulations in Section 5.2 provide an example the utility of the latent class structure of the linking model. The latent class structure provides the potential for distribution in distance to vary across matched pairs. If such a structure is unnecessary in an application, the sampler tends to collapse to using single class to estimate $\mathbf{\Sigma}$.

6 Linking the IAC and CMF

In this section, we describe the process of linking IAC and CMF records. We created a linked data product using records from the 2007 CMF, but the approach could be applied to other years to facilitate additional research. For simplicity of notation, unless otherwise noted we let CMF denote all records in the 2007 CMF records and IAC denote all 2007/2008 IAC records. In Section 6.1, we describe the data processing performed prior to linking the data. In Section 6.2, we describe the links resulting from applying LFCMV. In Section 6.3, we perform a post-linking analysis exploring energy efficiency.

6.1 Data cleaning, blocking and transformation

The purpose of linking in this application is to identify which records in the CMF received IAC energy assessments. However, the IAC and CMF records were collected by two different organizations. Some variables common to both files are recorded using an organization specific scale, variable name, and accuracy level. For instance, in the IAC, NAICS code is recorded as a 6-digit identifier, while in the CMF, up to 9-digits are recorded for each plant. Because of these discrepancies, pre-processing is required to convert the IAC and CMF records into the subsets F_1 and F_2, respectively, used for linking. We use F_1 to denote the subset of IAC records used for linking, while F_2 denotes the subset of CMF records considered as possible matches for F_1.

6.1.1 BVs and MVs

To create F_1 and F_2, we begin by examining the BVs used to block the records. As in the simulations in Section 4, we use the BV fields of state and NAICS code. In the CMF, the state where an plant is located is denoted using the standard notation of two upper case letters. A small number of records are missing the variable for state, and we exclude these records from consideration for F_2. In the IAC, however, the BV "state" is recorded with a variety of upper-lower case combinations. For instance, "AL","Al" and "al" all refer to the state Alabama. We convert the IAC labels to match the double capital format in the CMF, i.e., for plants referring to Alabama, state is recorded as "AL". Once the state

Table 5: Description of the matching variables used to link the data in Section 6. The first column gives the name of the variable in IAC database. The second columns lists the name of the associated variable in the CMF, with the abbreviation for that variable in the third column.

IAC Variable	**CM Variable**	**Abbr.**
Sales (in $)	Total Value of Shipments (in thousands of $)	TVS
Number of Employees	Total Employment	TE

variable is standardized, we create a list of all states associated with plants in the IAC. We then exclude from F_2 all CMF records from states not on this list. In addition to records pertaining to the 50 states, the IAC contains 11 records from plants in Puerto Rico. As there are no records corresponding to Puerto Rican plants in the CMF, these records are also excluded from F_1.

After blocking on state, we proceed to the BV of NAICS code. For each record in the IAC, we create a list of CMF records pertaining to the same 6-digit NAICS code and state as the IAC record. However, such direct blocking leads to a number of IAC records with no possible matches in the CMF. The IAC records are a subset of the CMF, and as such all records in the IAC should have a match in the CMF. The lack of matches for some records after direct blocking suggests that some of the NAICS codes in IAC or CMF may have some uncertainty. To account for this, we considering blocking on three different levels of NAICS agreement. Specifically, we create subsets with agreement on 4-digit, 5-digit and 6-digit NAICS codes. We refer to these as levels of NAICS blocking.

Let $F_1^{(x)}$ denote the subset of IAC records that have a non-empty block $F_{2(i)}^{(x)}$ when blocking is performed on state and x-digit NAICS code. Here, $x = 4, 5, 6$. Similarly, let $F_2^{(x)}$ denote the set of possible matches for $F_1^{(x)}$ when blocking on state and x-digit NAICS. Recall that NAICS codes are nested up to the 4-digit level, meaning that every record in $F_2^{(6)}$ will also be in $F_2^{(5)}$ and $F_2^{(4)}$. By the same logic, $F_2^{(4)}$ is the largest subset of CMF records to be linked, 5-digit blocking restricts this linking to a smaller subset $F_2^{(5)}$, and 6-digit blocking yields the smallest subset $F_2^{(6)}$.

Table 6: The number of IAC records to be matched at each level of NAICS blocking, i.e., the cardinality of $F_1^{(x)}$ for $x = 4, 5, 6$. There are 786 original IAC records from 2007/2008, excluding records for plants in Puerto Rico.

NAICS Level	4-digit	5-digit	6-digit
IAC Records	550	480	330

Once records are moved into blocks defined by state and x-digit NAICS, MVs are used to determine which records in $F_{2(i)}^{(x)}$ are a match for each i in $F_1^{(x)}$. The MVs are listed in Table 5, and we use the abbreviations from the CMF to denote the MVs. The scale of the TVS variable is different in the IAC than it is in the CMF. Specifically, the CMF variable TVS is recorded in thousands of dollars while the IAC variable is recorded in dollars. We divide the IAC value by 1000 to match the scale of the CMF variable.

6.1.2 Agreement filters

Blocking with respect to 4-, 5- and 6-digit NAICS all lead to some large blocks. To reduce the block size, we apply what we call an agreement filter to narrow the possible options for each IAC record. Let TVS_{fi} denote the recorded value for TVS in file f for record i, and similarly let TE_{fi} denote the recorded value for employment. For each i in $F_1^{(x)}$ and each $i^{'}$ in $F_{2(i)}^{(x)}$, we compute

$$d_1(i, i^{'}) = \frac{TVS_{1i} - TVS_{2i^{'}}}{TVS_{1i}}, \tag{30}$$

and

$$d_2(i, i^{'}) = \frac{TE_{1i} - TE_{2i^{'}}}{TE_{1i}}. \tag{31}$$

Here, $d_1(i, i^{'})$ and $d_2(i, i^{'})$ are the percent difference between IAC record i and the observed Census record $i^{'}$ on TVS and TE, respectively. If either (30) or (31) is greater than .3, we declare i and $i^{'}$ a non-link and remove $i^{'}$ from $F_{2(i)}^{(x)}$. Otherwise, we allow i and $i^{'}$ to be possible links. While this agreement filter does reduce the block sizes, it also results in some IAC records which have no possible matches in the CMF. We drop these IAC records from F_1. Table 6 shows the resulting size of F_1 at each level of NAICS blocking.

The agreement filter chosen here reflects a belief that for true matches, the values of TVS and TE in the IAC and CMF should be similar. The threshold of 30% allows some room for noise, recording error and time variation. In general applications, the choice to apply an agreement filter, as well as the choice of appropriate threshold, is at the discretion of the analyst.

6.1.3 Transformations and standardization

After applying the agreement filter described in Section 6.1.2, we have our final subsets $F_1^{(x)}$ and $F_2^{(x)}$ for each x. Before linking, we transform the MVs by taking the log. Taking the log puts the data on a similar scale, meaning that the mixture model should require fewer components to describe the distribution of the data. In a final pre-processing step, we standardize the MVs, as necessitated by the linking model in Section 2.3. The standardization is performed as described in Section 4.1. Let $j = 1$ refer to TVS and $j = 2$ refer to TE. Let $Y_{ij}^{(x)}$ denote the standardized value for MV j for record i in $F_1^{(x)}$. Similarly, let $X_{i'j}^{(x)}$ denote the standardized value for MV j for record i' in $F_2^{(x)}$.

6.2 Evaluating the links

We run LFCMV to link each pair of data sets $(F_1^{(x)}, F_2^{(x)})$ for $x = 4, 5, 6$. As described in Section 4, for $x = 4, 5, 6$, we initialize C by selecting, for each record i in $F_1^{(x)}$, the record i' in $F_{2(i)}^{(x)}$ such that

$$d(i, i'|x) = \sqrt{\sum_{j=1}^{2} \left(\frac{Y_{ij}^{(x)} - X_{i'j}^{(x)}}{Y_{ij}^{(x)}} \right)}, \tag{32}$$

is minimized. We then fit the LFCMV model using the posterior sampling steps described in Section 3.2.

Applying LFCMV results in 5000 post burn-in posterior estimates of C at each level of NAICS blocking. We use these estimates to compute posterior probabilities that each record $i' \in F_{2(i)}$ is a match to record i in $F_1^{(x)}$ by computing the proportion of posterior draws such that each i' in $F_{2(i)}^{(x)}$ is linked to its corresponding record i. For each i, we look at the posterior probabilities associated with possible matches i' in $F_{2(i)}^{(x)}$. If at least one of the potential matches in $F_{2(i)}^{(x)}$ has a posterior probability $> .5$, we say i has a high probability

Table 7: Displays the types of links obtained after linking at each level of NAICS blocking. We define a link as high probability if the posterior probability of linking (i, i') is greater than .5.

	Link Type		
NAICS Level	High Prob	Low Prob	Total
4-digit	320	230	550
5-digit	330	150	480
6-digit	270	60	330

match. Otherwise, we say i has low probability matches. Table 7 shows a summary of the counts of IAC records having high and low probability matches at each NAICS level. Counts have been rounded to satisfy disclosure protocols.

As we shift from one level of NAICS blocking to another, the subset of records $F_1^{(x)}$ and $F_2^{(x)}$ result in different possible links. For instance, a record in $F_1^{(4)}$ that has a possible match in $F_2^{(4)}$ may not be in $F_1^{(5)}$, i.e., may have any possible matches in the CMF when blocking on 5-digit NAICS. This can be seen in the difference in Tthe total column in Table 7. For instance, when shifting from 4-digit to 5-digit blocking, there are 70 records that, according to our choice of blocking and agreement filter, no longer have any possible matches in the CMF. Table 8 displays the number of IAC records which are lost, i.e., no longer have possible matches in the CMF, as the blocking level changes. Table 8 also displays rounded counts for the number of these lost records that correspond to high probability matches.

While some record links are lost as we compare results across different levels of NAICS blocking, some record links remain consistent. We use the posterior probabilities associated with each CMF record to determine which record in $F_2^{(x)}$ is the most likely match to each i in $F_1^{(x)}$. In other words, we determine which record in $F_2^{(x)}$ is the *highest* probability match. We note that these matches are not necessarily *high* probability matches. For some IAC records, the highest probability match remains consistent across different levels of NAICS blocking. Table 9 indicates the counts of such records at 4-digit to 5-digit NAICS blocking, as well as 5-digit to 6-digit blocking.

Table 8: Displays the types of links from IAC records that cannot be matched as we move from one level of NAICS blocking to a more restrictive blocking level. We define a link as high probability if the posterior probability of linking $(i, i^{'})$ is greater than .5.

	Link Type		
NAICS Change	High	Low	Total
4 to 5	60	10	70
5 to 6	110	40	150
4 to 6	150	70	220

Table 9: At each NAICS blocking level, for each i in $F_1^{(x)}$ there is a highest-probability match in $F_2^{(x)}$. The counts above indicate how many of these highest-probability matches stay the same across x-digit NAICS levels.

NAICS Level	4 to 5	5 to 6
Count	350	250

At each level of NAICS blocking, there are also some singleton IAC records, or records in $F_1^{(x)}$ that have exactly one option in $F_{2(i)}^{(x)}$. Similar to the seeds discussed in Section 2.3 in Dalzell and Reiter (2017), singleton pairs are linked independently of the process of estimating C; the pairs are declared links at every iteration. These singletons therefore inform the process of estimating $\mathbf{\Sigma}$, providing iteration-consistent links from which to estimate distance across matched pairs. Singletons are dependent both on x and the choice of agreement filter threshold. Because we have applied the agreement filter described in Section 6.1.2, all singletons are to be within 30% of the F_1 value for TVS and TE. Ideally, the relationship in the linked singleton pairs will be reflective of relationships present in the rest of the data. However, the latent class structure of LFCMV means that if the relationship in the singletons is different than the relationship across pairs in other blocks, the model can assign these records to their own latent class (or classes) with a reflective

covariance matrix. The number of singletons varies across x-digits NAICS blocking. To satisfy disclosure protocols, we do not provide counts of singletons at any level of NAICS blocking.

6.3 Post-linking analysis

The links estimated in Section 6.2 can be used to construct identifiers for the records in the CMF, specifying which records received an IAC energy assessment. In this section, we examine energy efficiency in plants that received IAC energy assessments versus those that did not. In addition to examining the data to see if there are year-specific trends, we are also interested in examining that trend across time. As the CMF is conducted every 5 years, we examine energy efficiency trends in 2007 and 2012. We do not make a causal argument about the relationship between receiving an assessment and energy efficiency, as many factors, including plant size, may impact the relationships observed in the data.

As a basis for our analysis, we begin with all CMF records from 2007 and 2012 that have a 4-digit NAICS code observed in the IAC. This limits our analysis to industries that were assessed in the time period of interest. We further limit our analysis sample to plants meeting IAC standards for assessment. IAC assessments are typically performed for plants with no more than 500 employees and with a gross annual sales of no more than $100 million (Muller, 2001, pg. 13). Let A_{2007} denote the set of CMF records from 2007 that satisfy the IAC restrictions for sales and employees and have 4-digit NAICS codes that appear in the IAC. Similarly, let A_{2012} denote the set of CMF records from 2012 that satisfy the IAC restrictions for sales and employees and have 4-digit NAICS codes that appear in the IAC. We refer to A_{2007} and A_{2012} as our analysis samples.

6.3.1 Energy metric

For each record $i^{'}$ in A_{2007} and A_{2012}, we compute an energy efficiency metric as follows:

$$EF_{i'} = log\left(\frac{TVS_{i'}}{CF_{i'} + EE_{i'}}\right). \tag{33}$$

Here, (33) represents the log of the inverse of the energy cost share. The denominator represents total energy cost and is computed by adding the cost of fuels (CF) to the electricity expenditure (EE).

Energy efficiency is known to vary across different industries, or, equivalently, across NAICS codes. Producing aluminum foil, for instance, has different energy requirements from producing potato chips. This suggests that rather than directly comparing values of (33) across plants, an analysis might be conducted at the NAICS specific level. However, there are not enough IAC-assessed plants within most NAICS codes to make such an analysis feasible. Instead, we adjust the metric in (33) so that it can be compared across all NAICS codes. To do this, we convert the efficiency metric in (33) into a standardized energy score as follows.

For each record $i^{'}$ in A_{2007}, we sort the records into groups defined by 6-digit NAICS codes. Within each group, we compute the mean and standard deviation of values of (33) for CMF records in the group. Let $\overline{EF}_g$ denote the group g specific mean, and similarly let $SD(EF)_g$ be the group specific standard deviation. For $EF_{i'}$ in group g let

$$\widetilde{EF}_{i'} = \frac{EF_{i'} - \overline{EF}_g}{SD(EF)_g}, \tag{34}$$

be the standardized value of $EF_{i'}$. We refer to the value of (34) as the energy score. The energy scores are on a common scale, describing how many standard deviation each plant's value of $EF_{i'}$ is from the group specific mean. This allows each corporation to receive an energy score relative to their peers as defined by 6-digit NAICS. For records in A_{2012}, we create energy scores for 2012 analogously.

6.3.2 Results

The records in A_{2007} and A_{2012} are divided into two categories. The first category, labeled "With Assessments", refers to plants in a given year that received an IAC assessment. The

remaining plants are categorized as "Without Assessments", meaning that these plants did not receive an IAC assessment in 2007 or 2008. The process of assigning records in A_{2007} and A_{2012} to these categories is dependent upon the linking structures estimated in Section 6.2. For each x-digit NAICS blocking level, and for each posterior sample $s = 1, \ldots, M = 5000$ of C, we create a binary vector $I^{(s)}$ of length a_{2007}, where a_{2007} denotes the number of records in A_{2007}. Here, a 1 in position $i^{'}$ means that record $i^{'}$ in A_{2007} is linked to an IAC record at iteration s; such plants are categorized as "With Assessment" records for iteration s. All other plants are categorized as "Without Assessment" records.

In order to compare the energy scores in 2007 and 2012, we also consider records in A_{2012}. Linking is performed on the 2007 records, meaning that the indicators for IAC assessments are associated with a 2007 CMF record. However, the Census Bureau uses a longitudinal identifier called SURVU_ ID that can be used to identify records for a given plant across time. For each iteration, we identify the set of SURVU_ ID numbers that, conditional on each $I^{(s)}$, received an energy assessment in 2007. The 2012 records corresponding to these unique identifiers compose the 2012 "With Assessment" samples.

Data cleaning is necessary before using SURVU_ ID as a longitudinal identifier. Exploration of the data revealed that SURVU_ ID is not unique within either the 2007 nor the 2012 CMF. This means multiple records with the same SURVU_ ID identifier exist within each year-specific data set. However, these records appear to be true duplicates, meaning every field value is the same. We delete all but one record with each SURVU_ ID to create a unique identifier.

For each iteration s and each level of NAICS blocking, we compute the density of energy scores for "With Assessment" plants, and separately for "Without Assessment" plants, for 2007 and 2012. We use fixed bins across the iterations. Within each year, and each of the two categories, we then average the density in each bin at each iteration and plot the results. The resultant average density curves for 2007 are displayed in Figures 8, 10 and 12, and the resultant averaged density curves for 2012 are displayed in Figures 9, 11 and 13.

In Figures 8 through 13, solid lines are used for the "Without Assessment" group and dashed lines are used for the "With Assessment" group. Each curve has a plotted 95%

Table 10: Displays 95% MI intervals for the mean of the "With Assessments" versus "Without Assessment" group at each level of NAICS blocking. Similar trends are evident across all three levels of NAICS blocking.

NAICS4		Mean	95% Interval
2007	With Assessments	-0.18	(-.27,-.09)
	Without Assessments	-0.007	(-0.01,-0.005)
2012	With Assessments	0.03	(-0.04,0.11)
	Without Assessments	0.008	(0.006,0.011)
NAICS5		Mean	95% Interval
2007	With Assessments	-0.17	(-0.26,-0.08)
	Without Assessments	-0.007	(-0.01,-0.005)
2012	With Assessments	0.02	(-0.06,0.10)
	Without Assessments	0.009	(0.006,0.011)
NAICS6		Mean	95% Interval
2007	With Assessments	-0.18	(-0.29,-0.08)
	Without Assessments	-0.007	(-0.009,0.005)
2012	With Assessments	0.04	(-0.05,.12)
	Without Assessments	0.009	(0.06,.011)

interval, representing the 95% interval for the density within each bin. The trends in the density curves are very similar across the three levels of NAICS blocking. In 2007, the density curve for the "With Assessment" group appears to be shifted further left than the "Without Assessment" curve. This suggests that the energy scores for assessed plants seem smaller than plants without assessments. In other words, relative to their peer group, plants receiving IAC audits appear less energy efficient. This trend appears to vanish in the plots corresponding to 2012. In Figures 9, 11 and 13, there is essentially no difference in the curves between the "Without" and "With Assessment" groups.

To supplement the graphical exploration of the data, we leverage the combining rules of Rubin (1987) to perform inference across the M estimated linkage structures at each

level of NAICS blocking. Using the "With" and "Without Assessment" groups described above, we compute the average energy score for each of the two groups for each of the M iterations. Using the combining rules of Rubin (1987), we compute 95% confidence intervals for the mean of each group. Results are displayed in Table 10 by NAICS blocking level. The intervals in the table suggest the same general trends observed in the density curves. In 2007, the average energy score tends to lower for the "With Assessment" groups, with non-overlapping intervals for the "With" and "Without" groups. This difference is much less pronounced in 2012, and the intervals overlap, suggesting no statistically significant difference in the means between the groups.

7 Conclusion

In this paper, we have developed an approach for file linking that incorporates some of the uncertainty from the matching into the linked data product. The methodology allows faulty continuous variables to be used for matching within a blocking structure defined by fault-free variables. We applied the method to create of a database with the potential to allow novel research to be conducted on the policy interventions of detailed energy-efficiency assessments to SME. By combining information from the Census Bureau with publicly available IAC records, we increase the utility of the IAC database by enabling longitudinal research, as well as cross sectional comparisons to SME that do not receive assessments. LFCMV results in multiple linked data sets, allowing any statistical inferences on the impact of this policy to account of the uncertainty in the linking itself.

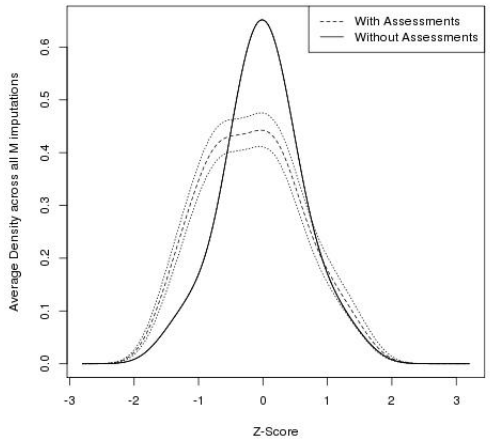


Figure 8: NAICS 4 Blocking, 2007: Kernel Density Plots. Solid line: Average Density of energy scores for CMF plants that did not receive assessments. 95% intervals are displayed but are very tight to the curve. Dashed lines: Average Density of energy scores for CMF plants that did receive assessments. 95% intervals are displayed. The fuzziness in the plot results from disclosure review requirements of the Census Bureau.

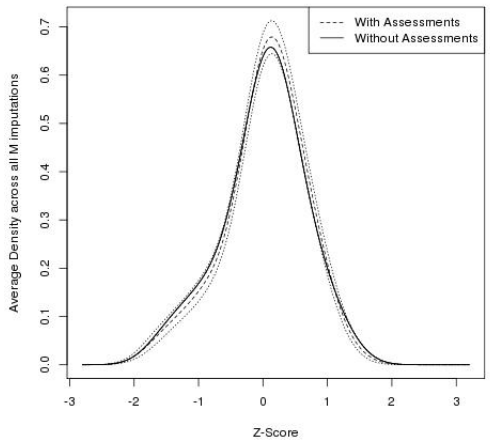


Figure 9: NAICS 4 Blocking, 2012: Kernel Density Plots. Solid line: Average Density of energy scores for CMF plants that did not receive assessments. 95% intervals are displayed but are very tight to the curve. Dashed lines: Average Density of energy scores for CMF plants that did receive assessments. 95% intervals are displayed. The fuzziness in the plot results from disclosure review requirements of the Census Bureau.

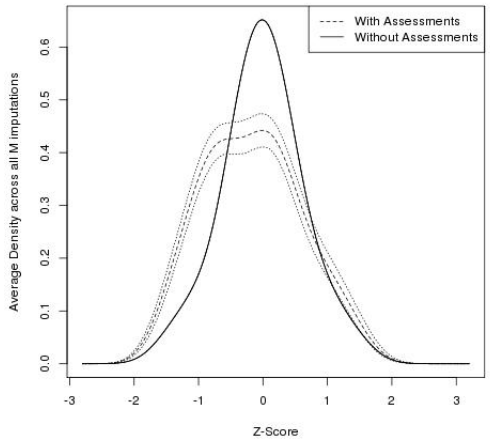


Figure 10: NAICS 5 Blocking, 2007 Kernel Density Plots. Solid line: Average Density of energy scores for CMF plants that did not receive assessments. 95% intervals are displayed but are very tight to the curve. Dashed lines: Average Density of energy scores for CMF plants that did receive assessments. 95% intervals are displayed. The fuzziness in the plot results from disclosure review requirements of the Census Bureau.

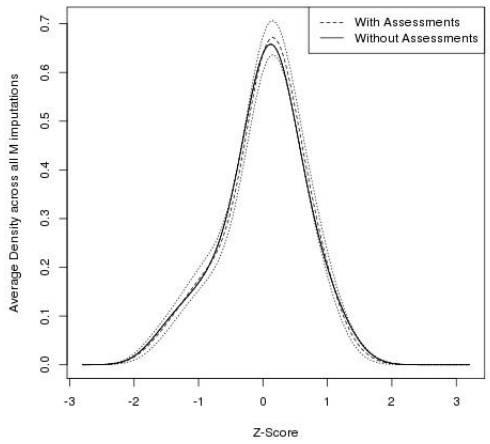


Figure 11: NAICS 5 Blocking, 2012: Kernel Density Plots. Solid line: Average Density of energy scores for CMF plants that did not receive assessments. 95% intervals are displayed but are very tight to the curve. Dashed lines: Average Density of energy scores for CMF plants that did receive assessments. 95% intervals are displayed. The fuzziness in the plot results from disclosure review requirements of the Census Bureau.

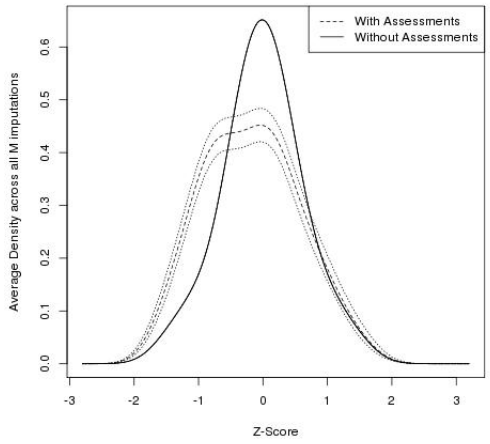


Figure 12: NAICS 6 Blocking, 2007: Kernel Density Plots. Solid line: Average Density of energy scores for CMF plants that did not receive assessments. 95% intervals are displayed but are very tight to the curve. Dashed lines: Average Density of energy scores for CMF plants that did receive assessments. 95% intervals are displayed. The fuzziness in the plot results from disclosure review requirements of the Census Bureau.

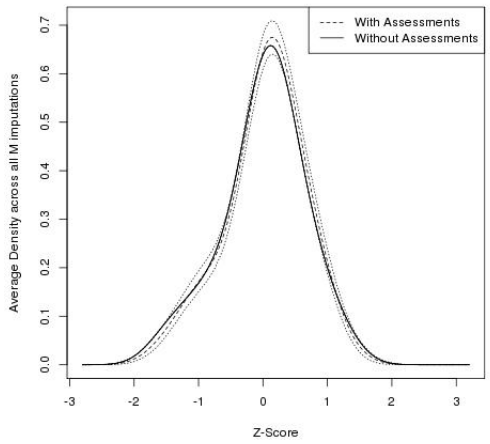


Figure 13: NAICS 6 Blocking, 2012: Kernel Density Plots. Solid line: Average Density of energy scores for CMF plants that did not receive assessments. 95% intervals are displayed but are very tight to the curve. Dashed lines: Average Density of energy scores for CMF plants that did receive assessments. 95% intervals are displayed. The fuzziness in the plot results from disclosure review requirements of the Census Bureau.

Acknowledgements

This work was supported by NSF Grant SES 1131897, and by the Duke University Energy Initiative Energy Research Seed Fund, with co-funding from the Information Initiative at Duke. Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed.

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